

GPU Programming with Directives

Overview



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Partners



Engineering and
Physical Sciences
Research Council

Natural
Environment
Research Council



THE UNIVERSITY
of EDINBURGH





- UK National Supercomputer Service
 - managed by UKRI/EPSRC
 - to be housed, operated and supported by EPCC
 - hardware Supplied by Cray
- Training provided by the ARCHER2 Computational Science and Engineering (CSE) support team
 - 60 days per year at various locations
 - free to all academics

What is EPCC?



- UK national supercomputer centre
 - founded in 1990 (originally Edinburgh Parallel Computing Centre)
 - a self-funding centre at The University of Edinburgh
 - running national parallel systems since Cray T3D in 1994
 - over 100 full-time staff
 - a range of academic research and commercial projects
- Get in contact if you want to collaborate
 - many staff are named RAs on research grants
 - joint research proposals
 - European project consortia
 - ...



Key ARCHER2 Resources

- Upcoming courses
 - <http://www.archer2.ac.uk/training/>
- Material from past courses
 - <https://www.archer2.ac.uk/training/materials/>
- Virtual tutorials (online)
 - <https://www.archer2.ac.uk/training/online/>
- Documentation
 - <https://www.archer2.ac.uk/documentation/>

Who are we?



David Henty d.henty@epcc.ed.ac.uk

Luca Parisi l.palisi@epcc.ed.ac.uk

Michael Bareford m.bareford@epcc.ed.ac.uk

Mark Bull m.bull@epcc.ed.ac.uk



Other Resources



- Please fill in the feedback form!
 - <https://www.archer2.ac.uk/training/feedback/>
- General enquiries about ARCHER go to the helpdesk
 - support@archer2.ac.uk
- EPCC runs one-year taught postgraduate masters (on-campus and online)
 - ***MSc in HPC*** and ***MSc in HPC with Data Science***
 - awarded by the University of Edinburgh since 2001
 - scholarships available
 - <http://www.epcc.ed.ac.uk/msc/>

Access to ARCHER2

- Your ARCHER2 ta229 project accounts
 - Small amount of budget
 - Other ARCHER2 accounts can be used
- Accounts will be closed a few weeks after access ends
 - all files etc. will be deleted
 - take copies of all your work beforehand!
- Course materials (slides, exercises etc) available from ARCHER2 website
 - archived on ARCHER2 web pages for future reference

Code of Conduct

<https://www.archer2.ac.uk/training/code-of-conduct/>

- We expect all course trainers and attendees to:
 - Use welcoming and inclusive language
 - Be respectful of different viewpoints and experiences
 - Gracefully accept constructive criticism
 - Focus on what is best for the community
 - Show courtesy and respect towards other community members
- See web page for full details and incident reporting form

Timetable

Session 1: Overview, GPU programming APIs, GPU Architecture

Session 2: Introduction to OpenMP, Data Movement

Session 3: Advanced Topics, Performance

Session 4: OpenACC, Practicals support

Session 5: Practicals support / Consultancy

Lecture notes etc.



Go to the ARCHER2 site at <https://www.archer2.ac.uk/> and follow the links to this course under the Training tab



Practical exercises source code



To download the source code for the practical exercises, make sure you are in your **work** directory on ARCHER2, then use the following command :

```
cp /work/z19/shared/GPUdir/*.tar .
```

I hope you enjoy the course



- ... and *please ask questions!*